

Sales Forecasting and Demand Analytics System

Executive Summary

This report presents the findings of a sales analysis and forecasting system developed to help improve business planning and inventory management. Historical sales data was analysed to understand customer purchasing behaviour, identify sales trends, predict future demand, detect unusual sales activities, and classify products according to their demand levels. The analysis shows that customer demand changes over time and varies across different product categories. A three-month sales forecast was prepared to support purchasing and inventory decisions. The system also identified unusual sales weeks that may require business attention and grouped products into demand-based categories so that inventory can be managed more efficiently. These insights can help the company reduce inventory costs, improve product availability, minimise stock shortages, and support better financial and operational decision-making.

1. Key Findings from EDA and Forecasting

Historical Sales Analysis

The analysis of historical sales data provided several important business insights:

- Sales are not constant throughout the year. Customer demand changes because of seasonal trends, holidays, promotional campaigns, and purchasing behaviour.
- Certain product categories consistently generate higher sales than others, indicating that customer demand is concentrated on a smaller number of high-performing products.
- Sales performance differs across regions, suggesting that inventory planning should consider regional demand rather than applying the same strategy everywhere.
- Some weeks experienced unusually high or unusually low sales, indicating that external business events or operational issues affected customer purchasing behaviour.

Forecasting Results

A forecasting model was developed to estimate future sales based on historical data.

The model achieved the following performance:

Performance Metric	Value
Mean Absolute Error (MAE)	15,281.24
Root Mean Square Error (RMSE)	18,524.63
Mean Absolute Percentage Error (MAPE)	21.36%

These results indicate that the forecasting model provides reasonably accurate predictions and can be used to support inventory planning and purchasing decisions.

2. Three-Month Sales Forecast with Confidence Ranges

The forecasting model estimated total sales for the next three months as follows:

Month	Forecasted Sales	Expected Sales Range
January 2019	45,942.74	18,855.87 – 73,029.61
February 2019	37,103.18	10,016.92 – 64,189.44
March 2019	69,034.93	41,950.30 – 96,119.56

Plain Language Explanation

The forecast predicts that sales are expected to remain healthy during the next three months, with March expected to generate the highest sales among the forecast period.

The confidence range represents the expected variation in sales. For example, January sales are predicted to be around **45,943**, but actual sales may reasonably

fall between **18,856** and **73,030** depending on customer demand, promotions, market conditions, or unexpected business events.

Rather than treating these numbers as exact values, management should use them as realistic planning estimates when making purchasing and inventory decisions.

3. Top Three Sales Anomalies and Their Likely Causes

The anomaly detection system identified **10 unusual sales weeks** during the historical analysis. These weeks showed sales behaviour that was significantly different from normal business operations.

Anomaly 1

Date: 22 March 2015

Sales: 37,703.67

Likely Cause

This unusually high sales value may have been caused by:

- Seasonal promotions
- Festival shopping
- Large customer orders
- Successful marketing campaigns

Anomaly 2

Date: 25 March 2018

Sales: 26,029.90

Likely Cause

Possible business reasons include:

- Limited-time discounts
- Increased customer demand before seasonal events
- Inventory clearance campaigns

Anomaly 3

Date: 8 October 2017

Sales: 28,412.10

Likely Cause

Possible reasons include:

- Holiday purchasing
- Bulk business purchases
- Regional promotional events
- Product launches

Business Importance

These unusual sales periods should be reviewed by management because they may indicate opportunities to increase future sales or reveal operational issues such as stock shortages, delayed deliveries, or unexpected demand changes.

4. Product Demand Segmentation and Stocking Strategy

Products were grouped into four demand segments based on their sales performance, sales growth, sales variability, and average order value.

Cluster 0 – Premium High-Value Products

Products:

- Copiers
- Machines

Characteristics:

- High average order value (**1,930.72**)
- Strong sales growth
- High sales volatility

Recommended Stocking Strategy

- Maintain moderate inventory levels.
 - Monitor demand frequently.
 - Avoid excessive stock because demand fluctuates significantly.
 - Focus on customer-specific purchasing requirements.
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Cluster 1 – Low Demand Products

Products:

- Appliances
- Art
- Bookcases
- Envelopes
- Fasteners
- Furnishings
- Labels
- Paper

Characteristics:

- Lowest sales volume
- Stable but slow demand
- Low average order value (**129.05**)

Recommended Stocking Strategy

- Maintain minimum inventory.
 - Purchase only when required.
 - Avoid overstocking to reduce storage costs.
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Cluster 2 – High Demand Products

Products:

- Accessories
- Binders
- Chairs
- Phones

- Storage
- Tables

Characteristics:

- Highest total sales (**239,495.78**)
- Consistent customer demand
- Moderate sales fluctuations

Recommended Stocking Strategy

- Maintain higher inventory levels.
 - Prioritize supplier relationships.
 - Replenish stock frequently.
 - Keep safety stock to prevent shortages.
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Cluster 3 – Fast Growing Niche Products

Products:

- Supplies

Characteristics:

- Highest sales growth rate (**1.93**)
- Moderate sales volume
- Demand increasing rapidly

Recommended Stocking Strategy

- Monitor demand closely.
- Gradually increase inventory.
- Review sales monthly to adjust stock levels.

5. Three Business Recommendations

Recommendation 1

Increase Inventory for High-Demand Products

Products in Cluster 2 generated the highest total sales (**239,495.78**).

Recommendation

Increase inventory for products such as Phones, Chairs, Storage, and Accessories to reduce stock shortages and improve customer satisfaction.

Recommendation 2

Monitor Sales Anomalies Regularly

The system detected **10 unusual sales weeks**, indicating periods of exceptional demand.

Recommendation

Investigate these periods to understand whether they were caused by promotions, seasonal demand, or operational issues. These insights can improve future sales planning.

Recommendation 3

Use Forecast Results for Purchasing Decisions

The forecasting model predicts approximately:

- January: **45,942.74**
- February: **37,103.18**
- March: **69,034.93**

Recommendation

Plan inventory purchases based on these forecasts to reduce unnecessary inventory costs while ensuring sufficient product availability during periods of higher demand.

6. Risk / Limitation of the System

Although the forecasting system provides useful business insights, it relies primarily on historical sales data. Unexpected events such as changes in customer preferences, economic conditions, supply chain disruptions, competitor activities, or major promotional campaigns may cause future sales to differ from the predicted values.

Therefore, management should use the forecast as a decision-support tool rather than treating it as an exact prediction. Forecasts should be reviewed regularly alongside current market conditions and updated whenever new sales data becomes available.

Conclusion

The Sales Forecasting and Demand Analytics System provides valuable insights that help management understand historical sales performance, estimate future demand, identify unusual sales behaviour, and optimise inventory management. The analysis shows that high-demand products should receive greater inventory attention, while slow-moving products should be stocked more conservatively. By combining sales forecasting, anomaly detection, and product demand segmentation, the company can improve operational efficiency, reduce inventory costs, minimise stock shortages, and make more informed business decisions. This system serves as an effective decision-support tool for both the Supply Chain department and the Finance department, enabling more accurate planning and improved business performance.